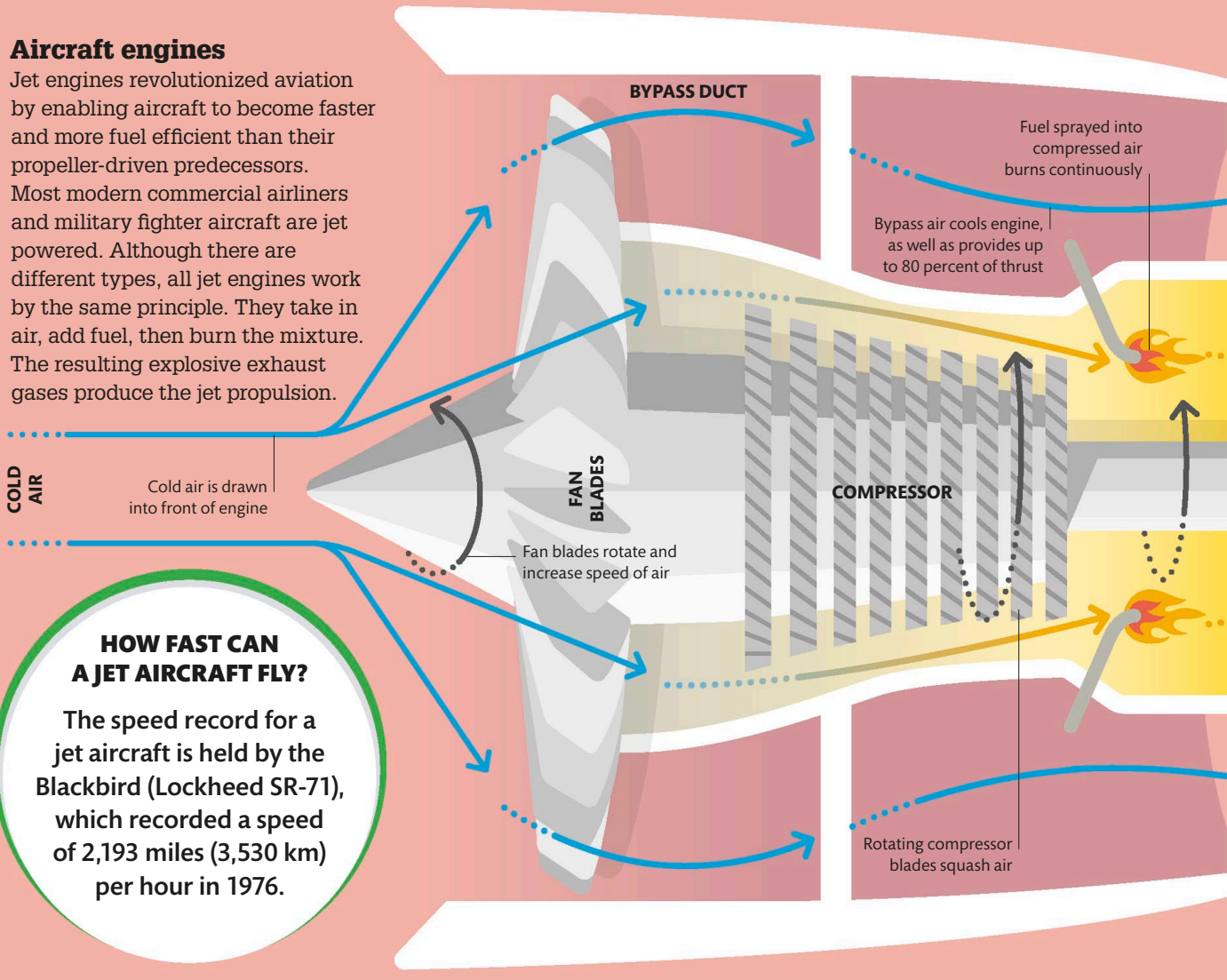
Jet engines and rockets

Jet engines and rockets are both types of reaction engines that use thrust to propel them forward or upward. The rapid expulsion of gas in one direction generates thrust in the opposite direction.

Turbofan engine
The most common type of jet engine used by passenger airliners is called a turbofan, named for the large fan at its front. In this type of engine, the main source of thrust is air that bypasses the central core.

Aircraft engines

Jet engines revolutionized aviation by enabling aircraft to become faster and more fuel efficient than their propeller-driven predecessors. Most modern commercial airliners and military fighter aircraft are jet powered. Although there are different types, all jet engines work by the same principle. They take in air, add fuel, then burn the mixture. The resulting explosive exhaust gases produce the jet propulsion.



HOW FAST CAN A JET AIRCRAFT FLY?

The speed record for a jet aircraft is held by the Blackbird (Lockheed SR-71), which recorded a speed of 2,193 miles (3,530 km) per hour in 1976.

1 Air intake
Fan blades at the front of the engine draw in cold air. Most of the air is propelled through bypass ducts to the back of the engine. The rest travels into the engine's core.

2 Compressor
The air enters the compressor, which contains a series of fan blades. This compresses the air, raising its temperature and pressure dramatically.

3 Combustion chamber
A steady stream of compressed air passes through to the combustion chamber. Here, fuel is sprayed in through nozzles, and the mixture burns at very high temperatures.